THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT

FORM THREE ANNUAL EXAMINATION
041 BASIC MATHEMATICS
MARKING SCHEME NOVEMBER 2025

SECTION A (60 MARKS)

1. (a) Let y *be* $0.\overline{135}$

$y = .0.\overline{135}$……..…….equation 1

$1\,000y = \times 0.\overline{135} \times 1\,000 = 135.\overline{135}$ …….equation 2

$$-\left|\begin{matrix} 1\,000y = 135.\overline{135} \\ y = \quad 0.\overline{135} \end{matrix}\right.$$

$999y = 135$……`01 mark.

$\frac{999y}{99} = \frac{135}{999}$

$= \frac{135}{999}$…….……01 mark.

$= \frac{135\div 27}{999\div 27} = \frac{5}{37}$……01 mark.

(b) $\frac{5}{100} \times 300\,000 = 15\,000$ ……..……01 mark.

$\frac{5}{100} \times 600\,000 = 210\,000$…….……..01 mark.

$210\,000 - 15\,000 = 195\,000$……..…….01 mark.

2. (a) $\left(\frac{3}{2}\right)^{n-1} = \left(\frac{3^3}{2^3}\right)^{n+3}$

$\left(\frac{3}{2}\right)^{n-1} = \left(\frac{3}{2}\right)^{3(n+3)}$

$\left(\frac{3}{2}\right)^{n-1} = \left(\frac{3}{2}\right)^{3n+9}$ ……01 mark.

$n - 1 = 3n + 9$……..01 mark.

$n - 3n = 9 + 1$

$-2n = 10$

$\frac{-2n}{-2} = \frac{10}{-2}$

$n = -5$ ……01 mark.

(b)$\log\left(\frac{764\times 36.56}{\sqrt{3\,845}}\right) = \log(764\ \times 36.56) - \log\sqrt{3\,845}$

$= \log(7.64\ \times 10^2) + \log(3.656\ \times 10^1) - \log(3.845\ \times 10^3)^{\frac{1}{2}}$

$= \log(7.64\ \times 10^2) + \log(3.656\ \times 10^1) - \frac{1}{2}(\log(3.845\ \times 10^3))$

$= \log 7.64 + 2 + \log 3.656 + 1 - \frac{1}{2}(\log 3.845 + 3)$

$= 2 + 1 + \log 7.64 + \log 3.656 - \frac{3}{2} - \frac{1}{2}\log 3.845$……..01 mark.

$= 3 - \frac{3}{2} + 0.8831 + 0.5630 - \frac{1}{2}(0.5849)$

$= 1.5 + 1.4461 - 0.29245$……..01 mark..

$= 2.9461 - 0.29245$

$= 2.65365$ ……..…..….01 mark.

3. (a) Let the number of women at Seypalm club be U
Let the number of women who own a land rover be A
Let the number of women who own a bicycle be B
$n(A) = 23, n(B) = 14,\ n(A \cap B) = 5$

$n(A \cup B) = n(A) + (B) - n(A \cap B)$……..……01 mark.

$= 23 + 14 - 5$
$= 37 - 5$…….01 mark.
$= 32$

Therefore, there are 32 women at Seypalm club…….……01 mark

(b) A= {5 , 10 , 15 , 20 , 25}…….00$\frac{1}{2}$ mark.

$B = \{10\ ,20\}$……..00$\frac{1}{2}$ mark.

$(A \cap B) = \{10\ ,20\}$……..01 mark.

$A \cup B = \{5\ , 10\ , 15, 20, 25\}$……..……01 mark

4. (a) $m = \frac{y_2 - y_1}{x_2 - x_1}$

$= \frac{4-2}{4+3} = \frac{2}{7}$……..……...01 mark.

$\frac{2}{7} = \frac{y-2}{x+3}$

$2x + 6 = 7y - 14$

$2x + 6 + 14 = 7y - 14 + 14$

$2x + 20 = 7y$

$\frac{7y}{7} = \frac{2x+20}{7}$ or $2x - 7y + 20 = 0$ ……..01 mark.

$y - intercept = c$ $from$ $y = mx + c$

Therefore $y - intercept = c = \frac{20}{7} or \left(0, \frac{20}{7}\right)$……01 mark.

(b) i. Corresponding angles are: d and h, e and a,
f and b, g and c……..…01 mark.

ii. Alternate interior angles are :
f and d , e and c…….……$00\frac{1}{2}$ mark.

iii. Alternate exterior angles are :
h and $b, g and$ a……..…..$00\frac{1}{2}$ mark.

iv. Vertically opposite angles are :
h and f, e and g, d and b, a and c……..…..01 mark.

5. (a) $\overline{AB} = \overline{AC}$ (given)……01 mark.
$\overline{BM} = \overline{CM}$ (Given)…….…….01 mark.
$\overline{AC}$ is common……..…..01 mark.
$\angle AMC = \angle BMA$ (By SSS- theorem)……..……01 mark.
$\overline{AM} \perp \overline{BC}$
Therefore $\angle AMC = \angle BMA = 90^{\circ}$…….……01 mark.

(b) $\frac{AO}{OD} = \frac{OB}{OC} = \frac{AB}{DC}$ = Constant of proportionality……..…01 mark.
$\frac{AO}{OD} = \frac{OB}{OC} = \frac{8\ dm}{4\ dm} = \frac{6\ dm}{3\ dm} = 2$……..…..…01 mark.
$\angle BOA = \angle DOC$ (Vertically opposite angle)…….01 mark.
$\Delta AOB \sim \Delta DOC$ (By SAS –theorem)…….……01 mark.

6. (a) $y \propto \frac{1}{x}$, $y = \frac{k}{x}$
$y_1 x_1 = k = x_2 y_2$

$k = 20 \times \frac{3}{2} = 30,$

$5y = 30, \frac{30}{5} = \frac{5y}{5}, y = 6,$…….01 mark.

$30 = 15y, \frac{30}{15} = \frac{15y}{15}, y = 2,$ …….01 mark.

$30 = 10y, \frac{30}{10} = \frac{10y}{10}, y =$

3……..…..01 mark.

x	5	10	15	20
y	6	2	3	1.5

(b) Let the distance travelled be d *and* l *be* litres of diesel used

$d \propto l, d = kl$

$k = \frac{d_1}{l_1} = \frac{d_2}{l_2}$……..……01 mark.

$= \frac{480}{32} = 15$……..……01 mark

$15 = \frac{d \times 20}{20}, d = 15 \times 20 = 300\ km$ ……01 mark.

Therefore 20 litres of diesel will cover a distance of 300 km.

Alternatively

$480\ km = 32\ litres$

$y = 20\ litres$ ……01 mark.

$y = \frac{480\ km \times 20\ litres}{32\ litres}$……01 mark.

$= 300\ km$

Therefore 20 litres of diesel will cover a distance of 300 km…..01 mark.

7. (a) Total parts $= 2 + 3 + 5 = 10$ parts……..…01 mark.

Second proportional $= \frac{3}{10} \times 3\ 000\ 000$ shillings…….01 mark.

$= 900\ 000$ shillings…….01 mark.

(b) CASH ACCOUNT (1)

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Date	Particular	Folio	Amount	Date	Particular	Folio	Amount
Jan 1	Capital	1	500 000	Jan 2	Machines	4	150 000
Jan 18	Seypalm	2	3 000 000	Jan 5	Purchases	5	2 000 000
	Sales	3	120 000				
				Jan 30	Balance	c/d	1 470 000
			3 620 000				3 620 000

Feb 1	Balance	b/d	1 470 000				

……03 marks.

8. (a) let the numbers be x , y and z

$18 , x, y , z , 30$

$A_1 = 18,\ A_2 = A_1 + d = x,\ A_3 = A_1 + 2d = y , A_4 = A_1 + 3d = z$

$A_5 = A_1 + 4d = 30$

$18 + 4d = 30$

$4d = 30 - 18$

$4d = 12$

$\frac{4d}{4} = \frac{12}{4}, d = 3$……01 mark.

$x = 18 + 3 = 21$ and……..……..01 mark.

$y = 18 + 2 \times 3 = 18 = 6 = 24$ ……$00\frac{1}{2}$ mark.

$z = 18 + 3 \times 3 = 18 + 9 = 27$……..…$00\frac{1}{2}$ mark.

Therefore, the numbers are 21, 24, and 27.

(b) $G_1 = x, G_2 = x, G_3 = x + 3$

$\frac{G_2}{G_1} = \frac{G_3}{G_2} = r$

$\frac{x}{x-2} = \frac{x+3}{x}$

$x^2 = x^2 + 3x - 2x - 6$

$x - 6 = 0, x = 6$……..…..……01 mark.

$G_1 = x - 2 = 6 - 2 = 4, G_2 = x = 6, G_3 = x + 3 = 6 + 3 = 9$

$r = \frac{G_2}{G_1} = \frac{6}{4} = \frac{3}{2}$…….……..01 mark.

$S_6 = G_1\left(\frac{r^n - 1}{r - 1}\right), r > 1$

$= \frac{4\left(\left(\frac{3}{2}\right)^6 - 1\right)}{\frac{3}{2} - 1}$

$= 8((1.5)^6 - 1)$

$= 83.125\ or\ 83\frac{1}{8}$……01 mark.

9. (a) $= \frac{\sqrt{3}+1}{1-\sqrt{3}}$……..01 mark.

$= \frac{\sqrt{3}+1}{1-\sqrt{3}} \times \frac{\sqrt{3}+1}{1+\sqrt{3}}$……$00\frac{1}{2}$ mark.

$= \frac{3+\sqrt{3}+\sqrt{3}+1}{(1)^2-(\sqrt{3})^2}$

$= \frac{4+2\sqrt{3}}{1-3}$

$= \frac{4+2\sqrt{3}}{-2}$……..…$00\frac{1}{2}$ mark.

$= \frac{-2-\sqrt{3}}{1}$

$= -1(2+\sqrt{3})$……..……01 mark.

$x + 4 = \sqrt{10^2 - 6^2}$

$x + 4 = \sqrt{100 - 36}$

$x + 4 = \sqrt{64}$

$x + 4 = 8$ ……..…..……..01 mark.

$x = 8 - 4, x = 4\ cm$……..…01 mark.

$y = \sqrt{6^2 + 4^2} = \sqrt{36 + 16} = \sqrt{52} = 7.2\ cm$…….01 mark.

10. (a)$2(2x + 3) - 10 < 6(x - 2)$

$4x + 6 - 10 < 6x - 12$

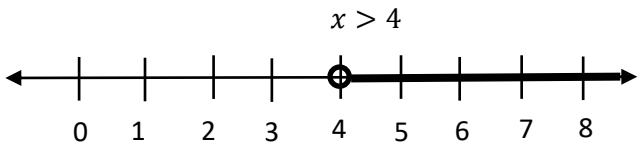
$4x - 4 < 6x - 12$

$4x - 6x < -12 + 4$

$-2x < -8$ ……..….01 mark.

$\frac{-2x}{-2} > \frac{-8}{-2}$

$x > 4$……..01 mark.



2 marks

(b) Let the numbers be $x\ and\ x + 2$

$x(x + 2) = 195$

$x^2 + 2x - 195 = 0$ ……01 mark.

$x = \frac{-2\pm\sqrt{4-4\times1\times-195}}{2}$

$= \frac{-2\pm\sqrt{4+780}}{2}$

$= \frac{-2\pm\sqrt{784}}{2}$

$= \frac{-2\pm28}{2}$……..01 mark.

$x = \frac{26}{2}\ or\ x = \frac{-30}{2}$

$x = 13 \; or \; x = -15$

$x = 13, x > 0$

$x = 12, x = 12 + 3 = 15$

Therefore, the numbers are 13 and 15……..……..……..01 mark.

SECTION B (40 MARKS)

11. (a) $\overline{AB} = \overline{BC}$ (Tangents from a common external point)……01 mark.

$CAB\angle = \angle BCA$ (Base angles of isosceles triangle)…… $00\frac{1}{2}$ mark.

$72^\circ + 72^\circ + a = 180^\circ$ (sum of interior angles of a triangle)……..…. $00\frac{1}{2}$ mark.

$a = 180^\circ - 144^\circ$

$a = 36^\circ$……..…..01 mark.

(b) Frequency distribution table

Class intervals	Frequency	x	fx	cf
$40 - 44$	7	42	294	7
$40 - 44$	8	47	376	15
$40 - 44$	11	52	572	26
$40 - 44$	10	57	570	36
$40 - 44$	4	62	248	40
$40 - 44$	2	67	234	42
Total	42	…..$00\frac{1}{2}$ mark.	$\Sigma fx_i = 2\,194$…01 mark.	

i. Mean $(\overline{x}) = \frac{\Sigma fx_i}{\Sigma f_i} = \frac{2\,194}{42}$……$00\frac{1}{2}$ mark.

$= 52.24$……..……..01 mark.

ii. Modal class is $50 - 54$

$Mode = l + \left(\frac{t_1}{t_1 + t_2}\right)c$

$= 49.5 + \left(\frac{11-3}{11-3+11-10}\right)5$

$= 49.5 + \left(\frac{3}{4}\right)5$……$00\frac{1}{2}$ mark.

$= 49.5 + \frac{15}{4}$

$= 49.5 + 3.75$

$= 53.25$.....01 mark.

iii. Median position $= \left(\frac{n}{2}\right)th = \frac{42}{2}th = 21th$

Median class is $50 - 54$

$$Median = l + \left(\frac{\frac{N}{2} - \Sigma fb}{fw}\right)c$$

$$= 49.5 + \left(\frac{\frac{42}{2} - 15}{11}\right)5$$

$$= 49.5 + \left(\frac{21-15}{11}\right)5$$

$$= 49.5 + \left(\frac{6}{11}\right)5$$ ……00$\frac{1}{2}$ mark.

$$= 49.5 + 2.7272 \ldots$$

$$= 52.23$$……01 mark.

12. (a) Angle $= \theta = (y - 5^{\circ})$

$l = \theta \times 60'$……00$\frac{1}{2}$ mark.

$= (y - 5^{\circ})60'$ Nautical miles……00$\frac{1}{2}$ mark.

$1\ nm = 1.852\ km$

$y = 2\ 500\ km$……00$\frac{1}{2}$ mark.

$y = \frac{1\ nm \times 2\ 500\ km}{1.852\ km}$

$= 1\ 349.89$ Nautical miles……01 mark.

$1\ 349.89 = (y - 5^{\circ}) \times 60'$

$\frac{1\ 349.89}{60} = \frac{(y-5^{\circ})60'}{60'}$

$22.498 = y - 5$ ……00$\frac{1}{2}$ mark.

$y = 22.498 + 5$

$= 27.498 \approx 27.5^{\circ}$……01 mark..

Therefore, the position of point Q is $27.5^{\circ}S, 30^{\circ}E$……..01 mark.

(b) Angle $= 43^{\circ} - 38^{\circ} = 5^{\circ}$……01 mark.

$l = \theta \times 60' \times \cos\theta$……00$\frac{1}{2}$ mark.

$= 5 \times 60' \times \cos 20^{\circ}$……00$\frac{1}{2}$ mark.

$= 281.91$ nautical miles……01 mark.

$1\ nm = 1.852\ km$

$281.91nm = y$……01 mark.

$y = \frac{1.852\ km \times 281.91\ nm}{1nm}$

$= 522.09\ km$……01 mark.

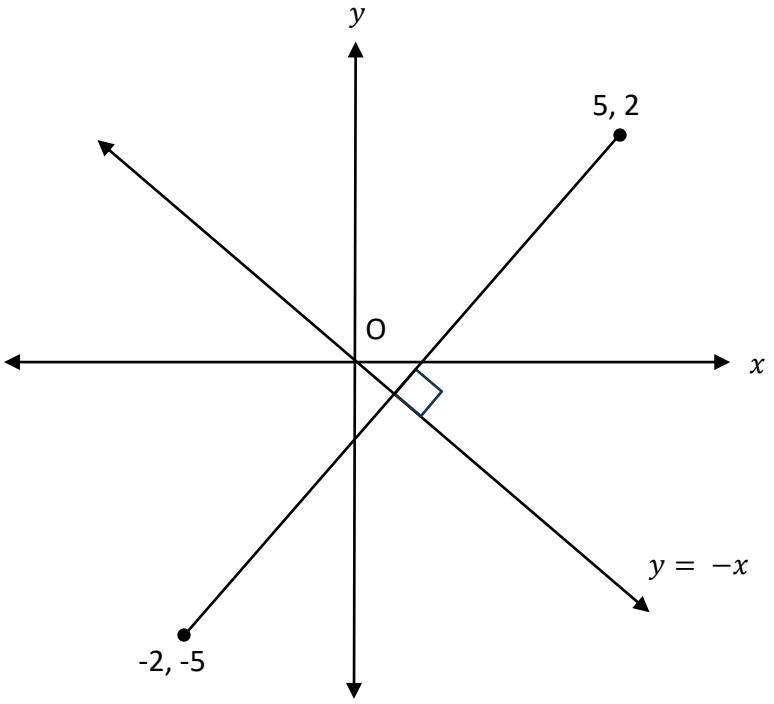
13. (a) $T(1\ unit\ left, 2\ units\ downward) = T(-1, -2)$……01 mark.

$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} a \\ b \end{pmatrix} + \begin{pmatrix} x \\ y \end{pmatrix}$……01 mark.

$= \begin{pmatrix} -1 \\ -2 \end{pmatrix} + \begin{pmatrix} 3 \\ 7 \end{pmatrix}$…….01 mark.

$= \begin{pmatrix} 2 \\ 5 \end{pmatrix}$……..…02 marks.

(b) $y = x = 0$ *is the same as* $y = -x$……..……01 mark.



02 marks

14. (a) $y = -3\ (x \leq 1)$

(b) ……..$00\frac{1}{2}$ mark.

x	...	-2	-1	0	1
y	…	-3	-3	-3	-3

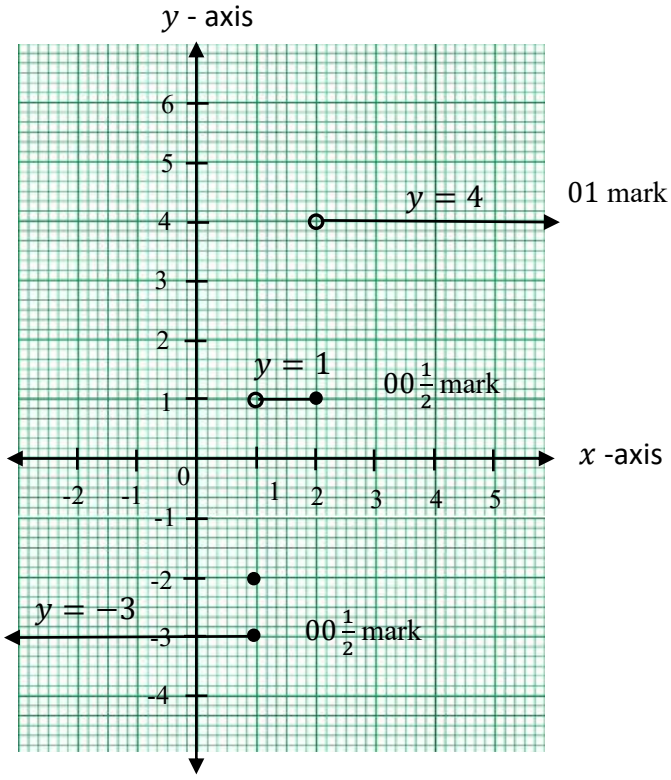
$y = 4\ (x > 2)$…… $00\frac{1}{2}$ mark.

x	2	3	4	5	….
y	4	4	4	4	…

$y - 1(1 < x \leq 2)$…… $00\frac{1}{2}$ mark

x	1	2
y	1	1

i.



∴

ii. Domain of $k(x) = \{x: x\mathcal{E}\mathbb{R}\}$……$00\frac{1}{2}$ mark.

Range of $k(x) = \{y: -3, 1, 4\}$……..…$00\frac{1}{2}$ mark .

iii. $k(-1) = -3$…….…….$00\frac{1}{2}$ mark.

(b)i. Domain of f $= \{x: x\mathcal{E}\mathbb{R}\}$…….…….01 mark.

Range of f= $y: y \geq \frac{4ac-b^2}{4a}$ …….……$00\frac{1}{2}$ mark

$= y: y \geq \frac{4\times1\times\times-3-(-2)^2}{4}$

$= y: y \geq \frac{-12-4}{4}$

$= \ y: y \geq \frac{-16}{4}$ ……..……$00\frac{1}{2}$ mark .

$= y: y \geq -4$…….…….…….….01 mark.

ii. Equation of symmetry

$x = \frac{-b}{2a} = \frac{--2}{2} = \frac{2}{2} = 1$……01 mark.

iii. Minimum value $= \frac{4ac-b^2}{4a} = \frac{4\times1\times-3-4}{4} = \frac{-12-4}{4} = \frac{-16}{4} = -4$……01 mark.